MPI

1. KRUŽNA LISTA, POSTAVLJANJE SREDNJE VREDNOSTI SEBE I DVA SUSSEDA

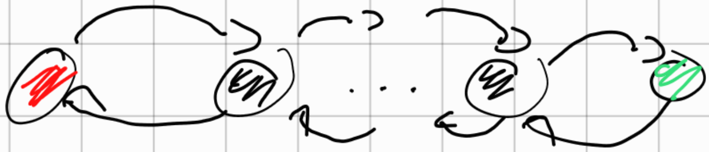
! PAZITI NA DEADLOCK (PARNI PRVO ŠALJU, NEPARNI PRVO PRIMAJU)

```
MPI_INIT()

ID = MPI_COMM_RANK()
m = MPI_COMM_SIZE()
O1,O2
IF( ID%2 ==0){
    SEND(VAL,(ID+1)%m)
    SEND(VAL,(ID-1)%m)
    RECV(O1,(ID+1)%m)
    RECV(O2,(ID-1)%m)
}ELSE{
    RECV(O1,(ID-1)%m)
    RECV(O2,(ID+1)%m)
    SEND(VAL,(ID-1)%m)
    SEND(VAL,(ID+1)%m)
}
MJ_NEW=(VAL+O1+O2)/3

MPI_FINALIZE()
```

2. ODREDITI MAX ELEM. U LANCU S 2 PROLAZA



```
MPI_INIT()

ID = MPI_COMM_RANK()
n = MPI_COMM_SIZE()
max = -INF

IF ( ID==0 )
    SEND ( VAL , ID+1 )
    RECV ( MAX, ID+1 )
ELSE IF ( ID== n-1 )
    RECV ( MAX , ID-1 )
    IF ( VAL > MAX )
        MAX=VAL
    SEND ( MAX , ID-1 )
ELSE
    RECV ( MAX , ID-1 )
    IF ( VAL > MAX )
        MAX =VAL
    SEND (MAX, ID+1 )
    RECV (MAX, ID+1 )
    SEND (MAX, ID-1 )
}
```

3. SIMULIRATI MPI_BARRIER

```
MPI_INIT()

ID = MPI_COMM_RANK(WORLD)
n = MPI_COMM_SIZE (WORLD)

IF (ID == 0)
    FOR(I IN (0, n-2))
        RECV(nullptr, *)
    FOR(I IN (1, n-1))
        SEND("1", i)
ELSE
    SEND("1", 0)
    RECV(nullptr, 0)

MPI_FINALIZE()
```

4. ANALIZA PROGRAMA SA SLIKE

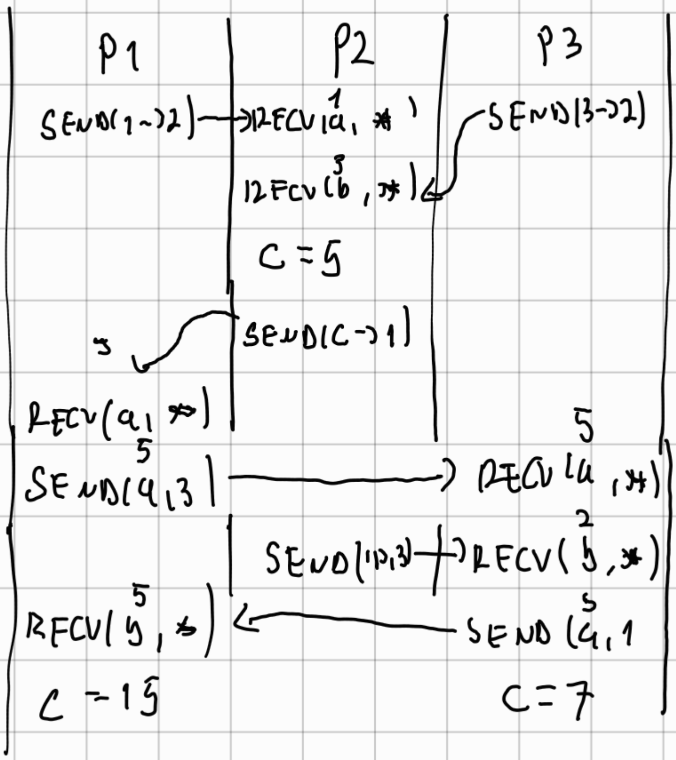
1°

P1	P2	P3
SEND(1->2)	RECV($\overset{1}{a}$, *)	SEND(3->2)
	RECV($\overset{3}{b}$, *)	
	C = 5	
	SEND(C->1)	
$\overset{5}{}$RECV(a, *)	SEND($\overset{2}{ID}$->3)	RECV($\overset{2}{a}$, *)
SEND($\overset{5}{a}$, 3)		RECV($\overset{5}{b}$, *)
RECV(b, *)		SEND($\overset{2}{a}$, *)
C = 12		C = 7

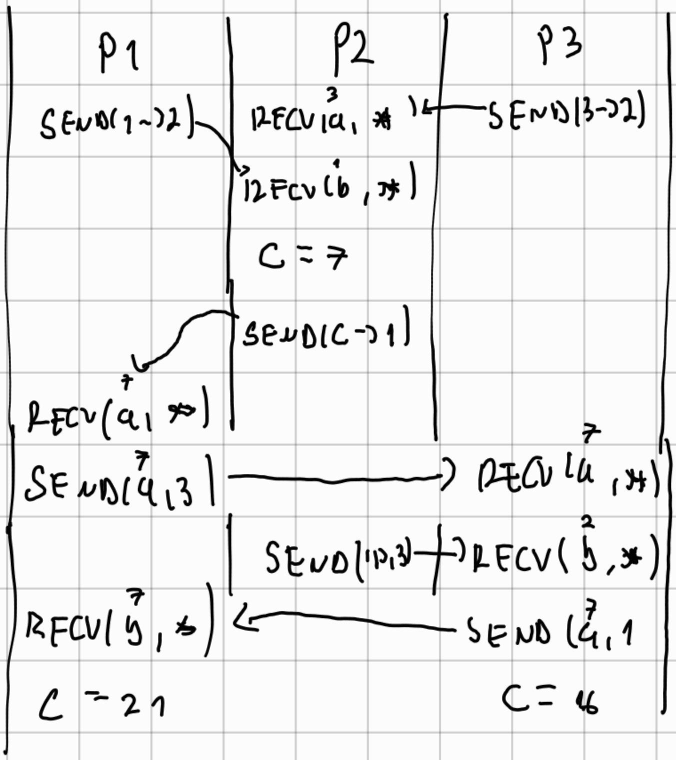
2°

P1	P2	P3
SEND(1->2)	RECV($\overset{3}{a}$, *)	SEND(3->2)
	RECV($\overset{1}{b}$, *)	
	C = 7	
RECV($\overset{7}{a}$, *)	SEND(C->1)	
	SEND(ID->3)	RECV($\overset{2}{a}$, *)
SEND($\overset{7}{a}$, 3)		RECV($\overset{7}{b}$, *)
RECV(b, *)		SEND($\overset{2}{a}$, *)
b = 16		

3°



4°



5. - IZRAČUNATI MINIMUM VARIJABLI U $\log_m$ DA SVI ZNAJU REZ => HIPERKOCKA

```
MPI_INIT()

ID = MPI_COMM_RANK(WORLD)
m= MPI_COMM_SIZE()
MIN=X
FOR( i in RANGE(0, log m - 1)){
    DEST= ID XOR 2^i
    SEND(X, DEST)
    RECV(X_O, DEST)
    IF XO < MIN{ MIN = X_O}
}
```

-> SVI IMAJU ISTI MINIMUM U VARIJABLI MIN

```
MPI_FINALIZE()
```

6. PROVERA DOGAĐAJA I BROADCAST SUMA

```
MPI_INIT()

ID = MPI_COMM_RANK(WORLD)
m = MPI_COMM_SIZE(WORLD)
INDEX = -1

IF(DOGAĐAJ)
    INDEKS = ID   -> JEDAN PROC IMA DOGAĐAJ

FOR (i; i < log m; i++) {
    DEST = ID XOR 2^i
    SEND( INDEKS , DEST )
    RECV( INDEKS_O , DEST )

    IF( INDEKS_O > -1)
        INDEKS = INDEKS_O
}
MPI_FINALIZE()
```

7. KRITIČNI ODSJEČAK - SAZNATI PRETHODNIKA I SLJEDBENIKA

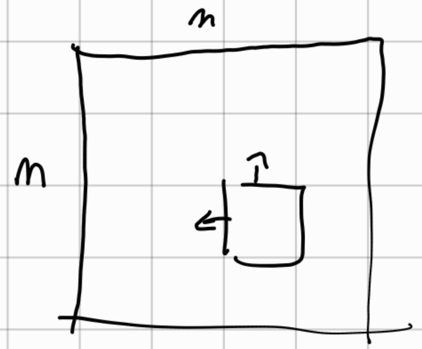
```
MPI_INIT()
ID = MPI_COMM_RANK(WORLD)
m = MPI_COMM_SIZE(WORLD)
P[] = [m] / OSTALI NA -1
P[KO] = ID
FOR (i; i < log m; i++) {
    DEST = ID XOR 2^i
    SEND( P , DEST )
    RECV( P_OTHER , DEST )
```

```
    FOR ( i ; i < n ) {
        IF ( P_OTHER[i] != -1 )
            P[i] = P_OTHER[i]
    }
}
```

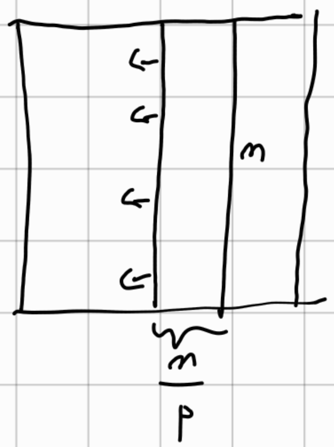
MPI_FINALIZE()

ANALIZA

(1.)



a) PODJELA NA p STUPACA



→ TREBAMO KOMUNICIRATI IZMEĐU STUPAC

$T_r = n^2 \cdot t_c$

$T_k = p \cdot 1 \cdot (t_s + n \cdot t_w)$

$T_c = 0$

$E = \frac{n^2 \cdot t}{n^2 \cdot t_c + p\,(t_s + n \cdot t_w)} \quad : \frac{1}{n^2}$

$p \sim n \quad = \frac{t}{t_c + \frac{t_s}{p} + t_w}$